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A Lagrangian Relaxation Heuristic for Capacitated Facility Location with Single-Source Constraints

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Facility location models are applicable to problems in many diverse areas, such as distribution systems and communication networks. In capacitated facility location problems, a number of facilities with given capacities must be chosen from among a set of possible facility locations and then customers assigned to them. We describe a Lagrangian relaxation heuristic algorithm for capacitated problems in which each customer is served by a single facility. By relaxing the capacity constraints, the uncapacitated facility location problem is obtained as a subproblem and solved by the well-known dual ascent algorithm. The Lagrangian relaxations are complemented by an *add* heuristic, which is used to obtain an initial feasible solution. Further, a final adjustment heuristic is used to attempt to improve the best solution generated by the relaxations. Computational results are reported on examples generated from the Kuehn and Hamburger test problems.

INTRODUCTION

Facility location models are applicable to problems in distribution systems, communication networks and other fields. Models with uncapacitated facilities have been used successfully in many contexts and are a subject of continuing theoretical interest; recent surveys are given in Cornuejols *et al.*¹ and in Krarup and Pruzan.² However, many applications involve facilities that are inherently capacitated—e.g. warehouses with limited floor space or telecommunications switches with limited capacity.

Here we describe a heuristic algorithm for capacitated facility location problems with *single-source* constraints; i.e. each customer is assigned to exactly one facility. The algorithm is based on a Lagrangian relaxation technique that incorporates the capacity constraints into the objective function, resulting in an uncapacitated facility location problem as a subproblem. These subproblems are solved by the dual ascent algorithm of Erlenkotter.³ (Fisher⁴ has noted that Lagrangian relaxation subproblems have typically been problems solvable in polynomial or pseudo-polynomial time; the uncapacitated facility location problem, on the other hand, is not.) The Lagrangian relaxations are complemented by an *add* heuristic that adds facilities one at a time to a set of open facilities, and is used to find an initial feasible solution. Further, a final adjustment heuristic that adjusts customer assignments is used to attempt to improve the best feasible solution generated by the relaxations.

Neebe and Rao⁵ describe a branch-and-bound method for the capacitated problem with single-source constraints. They formulate the problem as a set-partitioning problem and obtain bounds from a linear programming relaxation. Barceló and Casanovas⁶ describe a heuristic Lagrangian algorithm in which the assignment constraints rather than the capacity constraints are relaxed.

Many other authors have considered the capacitated problem in which a customer can be served by one or more facilities. Geoffrion and McBride,⁷ Nauss,⁸ Christofides and Beasley⁹ and Bitran *et al.*¹⁰ consider Lagrangian relaxation techniques with relaxed assignment constraints. Guignard-Spielberg and Kim¹¹ consider an approach for this problem that incorporates additional surrogate constraints. Other methods are given by Van Roy,^{12,13} Guignard and Spielberg,¹⁴ Akinc and Khumawala,¹⁵ Davis and Ray,¹⁶ Sa¹⁷ and Ellwein and Gray.¹⁸ In addition, problems with nonlinear economies-of-scale are considered by Kelly and Khumawala,¹⁹ Khumawala and Kelly²⁰ and Soland.²¹

PROBLEM AND ALGORITHM

In single-source, capacitated facility location problems, a number of possible locations, $i = 1, 2, \dots, N$, are available, each with a limited capacity b_i . A number of customers, $j = 1, 2, \dots, M$, must be served by the facilities; each customer requires some amount a_j of capacity. It is required to choose some subset of the possible facilities to be open, and then to assign customers to these open facilities. The amount of capacity at each facility i that is used by customers must not exceed the limit b_i . The objective is to minimize total costs, which consist of fixed set-up costs F_i for opening each facility i and assignment costs c_{ij} for serving all the requirements of customer j from facility i . The problem can be formulated as:

$$\text{Minimize} \left[\sum_{i=1}^N F_i y_i + \sum_{i=1}^N \sum_{j=1}^M c_{ij} x_{ij} \right] \quad (1a)$$

subject to:

$$\sum_{j=1}^M a_j x_{ij} \leq b_i \quad \text{for } i = 1, \dots, N, \quad (1b)$$

$$\sum_{i=1}^N x_{ij} = 1 \quad \text{for } j = 1, \dots, M, \quad (1c)$$

$$x_{ij} \leq y_i \quad \text{for } i = 1, \dots, N \quad \text{and } j = 1, \dots, M, \quad (1d)$$

$$x_{ij}, y_i \in \{0, 1\} \quad \text{for } i = 1, \dots, N \quad \text{and } j = 1, \dots, M. \quad (1e)$$

Binary variables y_i indicate whether facility i is open ($y_i = 1$) or not ($y_i = 0$), and binary variables x_{ij} indicate whether customer j is assigned to facility i ($x_{ij} = 1$) or not ($x_{ij} = 0$). Constraints (1b) ensure that capacity restrictions are satisfied. Constraints (1c) ensure that each customer is assigned, and constraints (1d) ensure that assignments are made only to open facilities.

By associating non-negative multipliers λ_i with constraints (1b) and incorporating them into the objective function, we obtain the subproblem

$$\text{Minimize} \left[\sum_{i=1}^N F_i y_i + \sum_{i=1}^N \sum_{j=1}^M (c_{ij} + \lambda_i a_j) x_{ij} - \sum_{i=1}^N \lambda_i b_i \right] \quad (2)$$

subject to (1c), (1d) and (1e).

An iterative procedure can, therefore, be designed in which the multipliers are updated between successive solutions of the uncapacitated subproblem (2). We propose to use the dual ascent and dual adjustment procedures of Erlenkotter³ (without branch-and-bound) to generate feasible solutions to (2). Empirically, these procedures produce excellent solutions to (2) in a relatively small amount of computation time. In fact, in most cases, they generate an optimal solution to (2). Further, since it is a dual-based method, a lower bound (often equal to the optimal solution) is always available. Since $\lambda_i \geq 0$ for all i , the lower bound of (2) provides a lower bound to problem (1). The goal is to find values for the multipliers λ_i for which the solution to (2) is feasible for the capacitated problem (1) and close to the best lower bound found. Of course, there may not exist λ_i s for which the solution of (2) is feasible for (1). One such simple example is as follows: $N = 2$, $M = 3$, $b_1 = b_2 = 10$, $a_1 = a_2 = 4$, $a_3 = 7$, $c_{11} = c_{22} = c_{23} = 1$ and $c_{12} = c_{13} = c_{21} = 2$.

The Lagrangian relaxation procedure starts with $\lambda_i = 0$ for all i . After solving (2) for the first time, each of the λ_i s that corresponds to a violated constraint is increased to the minimum value required to cause some customer to switch to another open facility. Thereafter, our updating rule for λ_i is similar to other updating rules based on subgradient optimization (see, for example, Fisher⁴). Specifically, we use

$$\lambda_i \leftarrow \text{Max} \left[\lambda_i + \frac{w(Z^u - Z^l) \left(\sum_{j=1}^M a_j x_{ij} - b_i \right)}{\left[\sum_{i \in U} \left(\sum_{j=1}^M a_j x_{ij} - b_i \right)^2 \right]^{1/2}}, 0 \right] \quad i \in U \quad (3a)$$

$$\lambda_i \leftarrow \lambda_i \quad i \notin U \quad (3b)$$

where

U = set of indices corresponding to multipliers that are allowed to change at the present iteration;

w = a positive scalar; we have used $w = 0.25$ with success;

Z^u = the best-known upper bound for problem (1), initialized by an *add* heuristic;

Z^l = the best-known lower bound for problem (1), initialized by solving (2) with $\lambda_i = 0$ for all i .

If (3a) is applied to all multipliers at each iteration, it allows the possibility that one or more values of λ_i may oscillate. Therefore, if a λ_i value decreases in one iteration and then increases in the next, we 'mark' that value of λ_i and prevent it from decreasing again. Thus, the set U consists of indices corresponding to multipliers that are not marked and to marked multipliers associated with violated constraints. After a feasible solution to (1) is found in one of the Lagrangian subproblems, we remove the 'marks' and restart the iterations, thus allowing λ_i s to decrease again. If, however, the new feasible solution is worse than one found previously through (2), we stop updating the λ_i s and execute the final adjustment heuristic. The Lagrangian relaxation heuristic is now outlined.

Lagrangian relaxation heuristic algorithm

- Step 0: Solve (1) by the *add* heuristic described below and initialize Z^u . Solve (2) with $\lambda_i = 0$ for all i and initialize Z^l . If the solution of (2) is feasible for (1), stop. Otherwise, initialize the λ_i s for the violated constraints, as described above.
- Step 1: Solve (2) and update Z^l if necessary. If the solution is feasible for (1), update Z^u if necessary, and go to step (4). Otherwise, continue.
- Step 2: If (2) has been solved MAX times (we set MAX = 200) without providing another feasible solution to (1), go to step (5). Otherwise, continue.
- Step 3: Update the λ_i s according to (3), mark multipliers as necessary, and go to step (1).
- Step 4: If $Z^u/Z^l \leq 1 + \varepsilon$ (we set $\varepsilon = 0.001$) or if a better solution of (1) was previously found by (2), go to step (5). Otherwise, record the solution, unmark multipliers, and go to step (3).
- Step 5: If the Lagrangian relaxations have not found any feasible solutions for (1), stop. Otherwise, execute the final adjustment heuristic described below, and stop.

The heuristic to find an initial feasible solution is an *add* procedure which adds facilities one at a time to the set K of open facilities.

Initial add heuristic

Step 0: Start with $K = \phi$, and set $Z^u = \infty$.

Step 1: For each facility $i \notin K$, compute an estimate R_i of the savings that would result if i were added to K , as follows. Let

$$w_{ij} = \text{Max} \left\{ \text{Min}_{k \in K} (c_{kj} - c_{ij}), 0 \right\} \quad \text{and} \quad \Omega_i = \sum_j w_{ij}, \quad i \notin K.$$

Compute

$$R_i = \Omega_i \text{Min} \left\{ \frac{b_i}{\sum_{\{j|w_{ij}>0\}} a_j}, 1 \right\} - F_i, \quad i \notin K,$$

where the minimum multiplier adjusts savings to reflect limited capacities.

Step 2: Open facility i' that achieves the maximum R_i for $i \notin K$, and add i' to K .

Step 3: If $\sum_{i \in K} b_i < \sum_j a_j$, return to step (1). Otherwise, continue.

Step 4: For each customer j , compute the cost differential between its best and second best assignment in K . Order j in order of decreasing cost differential.

Step 5: For each j in order, assign it to the open facility with minimum assignment cost among those with sufficient remaining capacity. If, for some j , no feasible assignment is possible, return to step (1). Otherwise, continue.

Step 6: Any facility i with no customers assigned to it should be removed from K and not be re-entered in future iterations. If the total cost TC of the last feasible solution found is less than Z'' , set $Z'' = TC$ and return to step (1). Otherwise, stop.

In the Lagrangian relaxation subproblems, customer assignments are made on the basis of costs $c_{ij} + \lambda_i a_j$. However, the actual cost incurred is simply c_{ij} . Consider a feasible solution generated by a Lagrangian relaxation subproblem. It is possible that a customer j was assigned to a facility k , even though sufficient capacity was available at some facility i with $c_{ij} < c_{kj}$, if, in fact, $c_{ij} + \lambda_i a_j > c_{kj} + \lambda_k a_j$. The final adjustment heuristic identifies situations of this type and makes appropriate adjustments in the customer assignments.

Final adjustment heuristic

Step 0: Let the initial incumbent solution be the best feasible solution generated by the Lagrangian subproblems.

Step 1: For each customer j , compute the cost differential between its best and second best assignment in K . Order j in order of decreasing cost differential.

Step 2: For each j in order, examine alternative facility assignments. Reassign customer j if j can be feasibly moved to another open facility with lower assignment cost.

Step 3: If no reassignments were made in step (2), stop. Otherwise, repeat step (2).

More sophisticated post-processing heuristics can, of course, be designed to change customer assignments. These could also be extended to the event that the Lagrangian relaxation subproblems fail to find a feasible solution. Note, however, that even if the Lagrangian routine fails to attain feasibility, in general, a feasible solution is still available from the *add* heuristic used initially.

COMPUTATIONAL RESULTS

Some experiments were performed using the 12 test problems in Kuehn and Hamburger.²² In these problems, $N = 25$ or 26 and $M = 50$. Demands a_j were modified from values in Kuehn and Hamburger.²² Specifically, all demands in excess of 3700 were set equal to 3700. This would allow feasible single-source solutions with reasonable-sized facilities. Two different capacity values were tried. In one set of runs, all facilities were given capacities of 7000; in the other, they were given capacities of 6000.

The computer code was written in FORTRAN, compiled using a FORTRAN 77 compiler, and run on an AMDAHL 470/V8 computer. The results are summarized in Table 1. The first line for each example is for a capacity of 7000, and the second line is for a capacity of 6000.

Consider momentarily just the feasible solutions that were generated by Lagrangian subproblems and the final adjustment heuristic (ignoring the *add* heuristic). Only in one problem did the solution of (2) fail to provide a feasible solution for (1). Among the others, 22 out of 23 yielded final Lagrangian solutions within 10% of the known lower bound; all were within 12% of the lower

TABLE 1. Numerical examples

Problem	Lower bound	Solution with final adjustments†		Best Lagrangian solution†		First Lagrangian solution†		Add heuristic % over lower bound
		Total time (sec.)	% over lower bound	Total iterations	% over lower bound	Iteration when found	% over lower bound	
1	797,379	5.8	0.08*	30	0.11	9	0.39	0.14
	797,497	17.3	0.17*	62	0.17	18	0.41	0.34
2	854,678	28.2	1.96	67	2.53	34	2.53	0.68*
	854,678	6.5	3.05	31	4.42	30	4.42	2.99*
3	893,753	104.8	4.19	131	5.27	48	7.56	1.87*
	893,770	30.6	3.53*	114	3.57	32	9.41	7.43
4	934,829	42.7	6.06	78	11.78	37	11.78	5.24*
	932,492	30.2	8.67*	219	12.30	38	16.47	15.80
5	1,092,891	4.6	0.25*	25	0.27	12	0.27	0.72
	1,092,891	7.1	0.80	52	0.98	33	1.45	0.79*
6	1,149,186	7.4	1.17*	27	3.42	25	3.42	1.59
	1,150,989	12.6	1.88	42	3.45	41	3.46	1.62*
7	1,193,017	78.6	3.64	153	5.76	66	6.31	1.41*
	1,194,819	171.2	3.64	145	5.10	46	5.10	3.33*
8	1,244,963	156.1	3.88	216	5.58	105	5.58	1.99*
	1,245,040	114.7		200		Not found		6.99*
9	614,522	0.7	0.36*	11	0.67	6	0.67	18.65
	614,775	3.9	1.35*	36	1.63	8	1.78	19.11
10	659,957	2.9	2.00*	14	3.24	12	3.24	17.20
	659,957	3.6	4.82*	17	6.80	15	6.80	17.78
11	690,717	5.6	7.16*	21	7.86	18	8.75	17.05
	690,717	26.9	6.83*	49	7.21	16	9.89	19.05
12	724,857	3.6	8.45*	21	12.26	20	12.26	18.77
	724,857	5.0	11.09*	25	13.24	21	13.24	22.75

†Excluding the add heuristic.

bound, and the average was 4.0% of the lower bound. For 10 out of 23 problems, a better solution was obtained by ‘unmarking’ multipliers and allowing them to decrease after finding the first feasible Lagrangian solution. Note, however, that in 6 of the 23 problems, more than 100 iterations were needed for convergence.

Variations of the algorithm were also tried. These included:

- (i) implementing a branch-and-bound on top of the dual ascent algorithm to guarantee an optimal solution of the Lagrangian subproblem (2);
- (ii) using a variable value of the constant w which starts at $w = 2$ and is halved every 10 iterations;
- (iii) using an alternate formulation $\sum_{j=1}^M a_j x_{ij} \leq b_i y_i$ for constraints (1b); this changes the value of the fixed costs in the subproblems;
- (iv) allowing multipliers to oscillate twice before ‘marking’ them.

In all cases, no consistent pattern was detected. Each gave better results for some problems and worse for others. In no case was one of the variations consistently better or worse than the base runs in Table 1.

The performance of the initial add heuristic is also summarized in Table 1. The heuristic performed very well in problems (1)–(8), with an average solution of 3.3% above the lower bound. However, the heuristic performed rather poorly in problems (9)–(12), with an average of 18.8% above the lower bound. This is so because in problems (9)–(12), the assignment costs c_{ij} are relatively low. The heuristic assigns customers to facilities without combinatorial considerations and thus may open more facilities than warranted. The additional fixed costs incurred are not sufficiently compensated in problems (9)–(12) by assignment cost savings.

Therefore, although on some problems the add heuristic solution was marginally better than the final Lagrangian relaxation solution, on many other problems the Lagrangian relaxation solution was far better. Considering the Lagrangian relaxation heuristic algorithm as a package, we conclude that Lagrangian relaxation, complemented by the add heuristic and the final adjustment heuristic, performed very well. The best solutions found are marked in Table 1 by asterisks. In 7

problems, these solutions were within 1% of the lower bound, and in 14 problems within 3%. On average, the solution was 3.5% above the lower bound.

FINAL REMARKS

The Lagrangian relaxation heuristic algorithm performed very well. Moreover, it provides a readily available lower bound. Relaxing the capacity constraints is attractive because the dual ascent algorithm used to solve the resulting subproblems is very fast. Of course, the Lagrangian relaxation algorithm can be incorporated into a standard branch-and-bound scheme so that an optimal solution will always be found.

On average, the *add* heuristic performed quite well, but in certain problems it performed poorly, and thus should not be used as a stand-alone method. Other *add* heuristics with more complex rules can be designed to obtain an initial solution. Other possibilities include *drop* heuristics, in which one starts with all facilities open and thereafter closes facilities one at a time, and combinations of *add* and *drop* heuristics.

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REFERENCES

1. G. CORNUEJOLS, G. L. NEMHAUSER and L. A. WOLSEY (1983) The uncapacitated facility location problem. Management Science Research Report No. MSRR 493, Carnegie-Mellon University [to appear in *Discrete Location Theory* (R. L. FRANCIS and P. B. MIRCHANDANI, Eds). Wiley-Interscience, New York].
2. J. KRARUP and P. PRUZAN (1983) Simple plant location problem: survey and synthesis. *Eur. J. Opl Res.* **12**, 36–81.
3. D. ERLINKOTTER (1978) A dual-based procedure for uncapacitated facility location. *Opns Res.* **26**, 992–1009.
4. M. L. FISHER (1981) The Lagrangian relaxation method for solving integer programming problems. *Mgmt Sci.* **27**, 1–8.
5. A. W. NEEBE and M. R. RAO (1983) An algorithm for the fixed-charge assigning users to sources problem. *J. Opl Res. Soc.* **34**, 1107–1113.
6. J. BARCELÓ and J. CASANOVAS (1984) A heuristic Lagrangian algorithm for the capacitated plant location problem. *Eur. J. Opl Res.* **15**, 212–226.
7. A. M. GEOFFRION and R. MCBRIDE (1978) Lagrangian relaxation applied to capacitated facility location problems. *AIIE Trans.* **10**, 40–47.
8. R. M. NAUSS (1978) An improved algorithm for the capacitated facility location problem. *J. Opl Res. Soc.* **29**, 1195–1202.
9. N. CHRISTOFIDES and J. E. BEASLEY (1983) Extensions to a Lagrangian relaxation approach for the capacitated warehouse location problem. *Eur. J. Opl Res.* **12**, 19–28.
10. G. R. BITRAN, V. CHANDRU, D. E. SEMPOLINSKI and J. F. SHAPIRO (1981) Inverse optimization: an application to the capacitated plant problem. *Mgmt Sci.* **27**, 1120–1141.
11. M. GUIGNARD-SPIELBERG and S. KIM (1983) A strong Lagrangian relaxation for capacitated plant location problems. Department of Statistics Technical Report No. 56, The Wharton School, University of Pennsylvania.
12. T. J. VAN ROY (1980) A cross decomposition algorithm for capacitated facility location. Working Paper 80-8A, Katholieke Universiteit Leuven.
13. T. J. VAN ROY (1983) Cross decomposition for mixed integer programming. *Mathl Progr.* **25**, 46–63.
14. M. GUIGNARD and K. SPIELBERG (1979) A direct dual method for the mixed plant location problem with some side constraints. *Mathl Progr.* **17**, 198–228.
15. U. AKINC and B. M. KHUMAWALA (1977) An efficient branch and bound algorithm for the capacitated warehouse location problem. *Mgmt Sci.* **23**, 585–594.
16. P. S. DAVIS and T. L. RAY (1972) A branch-bound algorithm for the capacitated facilities location problem. *Nav. Res. Logist. Q.* **19**, 753–759.
17. G. SA (1969) Branch and bound and approximate solutions to the capacitated plant location problem. *Opns Res.* **17**, 1005–1016.
18. L. B. ELLWEIN and P. GRAY (1971) Solving fixed charge location-allocation problems with capacity and configuration constraints. *AIIE Trans.* **3**, 290–298.
19. D. L. KELLY and B. M. KHUMAWALA (1982) Capacitated warehouse location with concave costs. *J. Opl Res. Soc.* **33**, 817–826.
20. B. M. KHUMAWALA and D. L. KELLY (1974) Warehouse location with concave costs. *INFOR* **12**, 55–65.
21. R. M. SOLAND (1974) Optimal plant location with concave costs. *Opns Res.* **22**, 373–385.
22. A. A. KUEHN and M. J. HAMBURGER (1963) A heuristic program for locating warehouses. *Mgmt Sci.* **9**, 643–666.